

A comparison of open source CM tools

	Platform support	Licence	Language	Website
Chef	AIX, *BSD, HP-UX, Linux, Mac OS X, Solaris, Windows, others	Apache	Ruby	https://www.chef.io/
Puppet	AIX, *BSD, HP-UX, Linux, Mac OS X (partial), Solaris, Windows, others	Apache	Ruby	http://puppetlabs.com/
CFEngine	AIX, *BSD, HP-UX, Linux, Mac OS X, Solaris, Windows, others	C	GPL, COSL	http://cfengine.com/
Juju	Linux	Python, Go	Affero General Public License	https://juju.ubuntu.com/
Ansible	AIX, *BSD, HP-UX, Linux, Mac OS X, Solaris, Windows, others	GPL	Python	http://www.ansible.com/
Salt	AIX, *BSD, HP-UX (partial), Linux, Mac OS X, Solaris, Windows, others	Apache	Python	http://www.saltstack.com/
Vagrant	Linux, Mac OS X, Windows	MIT License	Ruby	https://www.vagrantup.com/
Bcfg2	AIX (partial), *BSD, Linux, Mac OS X (partial), Solaris	BSD	Python	http://bcfg2.org/
Rudder	AIX, *BSD (partial), Linux, Mac OS X (partial), Solaris (partial), Windows, others	GPL and Affero GPL	C and Scala	http://www.rudder-project.org/
SmartFrog	HP-UX, Linux, Mac OS X, Solaris, Windows	Java	LGPL	http://www.smartfrog.org/display/st/SmartFrog+Home

Bcfg2 server:

- It accepts network descriptions and converts them into a series of configuration specifications
- It manages data provided by clients
- It keeps track of statistics for clients
- It takes information from a pool of metadata that describes clients, and from the file system repository for mappings from metadata to literal configuration

on the network. It creates components based on the SmartFrog system's descriptions, which is available in the Configuration Description Language. It enables interactions between components and lifecycle monitoring for executing tasks.

Modification in a configuration file can change the deployment architecture swiftly and, hence, it is easier to experiment and innovate. It also enables agility and scalability. 

SmartFrog

SmartFrog (Smart Framework for Object Groups) is an open source tool written in Java. It is used to manage configuration, deployment and orchestration of various distributed components that comprise a robust system. A prototype-based object-oriented language or domain-specific language is used for component configuration. It encapsulates systems in such a way that it is easy to manage components, to install them automatically and to de-provision.

SmartFrog consists of:

- Configuration Description Language:** This is used to define configurations, to provide system modelling capabilities, notions of abstraction and template creation, and notation for describing system configurations.
- Component Model:** This specifies how SmartFrog components can be written and integrated into the framework. It manages the lifecycle of components. It also supports dynamic discovery and the naming of components, distributed workflows, sequencing, script execution, installation and failure detection.
- Runtime System:** This consists of daemons that run

References

- http://en.wikipedia.org/wiki/Configuration_management
- http://en.wikipedia.org/wiki/Comparison_of_open-source_configuration_management_software
- http://www.tomsitpro.com/articles/open-source-cloud-computing-software_2-754-10.html
- <http://searchdatacenter.techtarget.com/definition/configuration-management-CM>
- http://en.wikipedia.org/wiki/Software_configuration_management
- http://www.sei.cmu.edu/productlines/frame_report/config_man.htm
- <https://juju.ubuntu.com/deployment/>
- <http://docs.bcfg2.org/reports/dynamic.html>

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